Automated Weekly Hotel Performance Reports

AI Management Trainee Assignment | Working n8n workflow, tested reference implementation, and design rationale

10 hotels tested | **5 doing well** | **5 need improvement**

Executive summary

The proposed system runs every Monday morning, reads the latest hotel performance rows from Google Sheets, validates the inputs, calculates transparent peer benchmarks, and classifies each hotel. It then asks an AI model to write a concise, factual email and sends it through Gmail. A deterministic fallback message keeps the batch moving if the model is unavailable or returns invalid output.

The supplied workbook is a single weekly snapshot with revenue, order count, hotel name, and owner email. It does not contain dates or hotel-specific goals. For this version, the scheduled run treats the sheet as the previous week and uses peer medians. This is intentionally simple, auditable, and appropriate for the available data.

Solution architecture

Why these tools

Tool	Role	Reason
n8n	Orchestration	Visible workflow, scheduling, credentials, retries, and easy handoff to operations.
Google Sheets	Source	Matches the assignment and lets business users maintain data without a deployment.
OpenAI Responses API	Message generation	Structured output produces separate subject and HTML body fields. The prompt is grounded only in calculated facts.
Gmail	Delivery	Direct owner communication with a controlled test-recipient mode before production.
Python	Reference test	Provides a reproducible local dry run and validates the complete decision logic without credentials.

Decision logic

The mock dataset produces a median weekly revenue of **285,500** and a median weekly order count of **62.5**. A hotel is classified as **Doing well** only when both metrics meet or exceed their median. If either metric is below median, it is classified as **Needs improvement**. Zero revenue and zero orders also trigger a no-activity warning.

Why the median: it is easy to explain, recalculates automatically as the portfolio changes, and is less distorted by unusually large hotels than an average. Why both metrics: it prevents high order volume with weak booking value, or high revenue from very few orders, from masking a problem.

Results from the supplied mock data

Hotel	Revenue	Orders	Classification
Songhe Inn	482,000	96	Doing well
Riverside Hotel	315,000	71	Doing well
Blue Bay Hostel	128,000	54	Needs improvement
Sky Spa Resort	598,000	102	Doing well
Orange Light	256,000	49	Needs improvement
Forest Hill	92,000	21	Needs improvement
Cape Cottage	0	0	Needs improvement
Starlight Biz	374,000	80	Doing well
Artis Inn	167,000	37	Needs improvement
Ocean View	442,000	88	Doing well

How this should evolve

Peer medians are a starting benchmark, not a permanent business target. The next data version should add weekly date, hotel-specific revenue goal, order goal, room inventory, and prior-week values. The classifier can then prioritize target attainment and trend, while retaining the peer median only as context.

Data-quality rules

- Reject rows with a missing hotel name, owner email, or non-numeric revenue/orders.
- Reject negative performance values and flag zero activity for data-feed review.
- Keep original metrics separate from calculated benchmarks for auditability.
- Redirect all recipients to one controlled inbox while test mode is enabled.

AI message design

The AI receives only one hotel's validated facts and the calculated portfolio benchmarks. It does not decide the classification. Keeping the business rule in code makes outcomes deterministic, testable, and consistent; the model is used only for clear, personalized wording.

Prompt contract

Input	Hotel name, week dates, revenue, orders, medians, status, and no-activity flag.
Restrictions	Use only supplied facts. Do not invent causes, goals, percentages, or comparisons.
Tone	Supportive, concise, specific, and professional.
Action	Include one practical next step appropriate to the status.
Output	Strict JSON with subject and safe simple HTML body.
Fallback	Use a deterministic template if the API fails or the output does not validate.

Example message

Subject: Weekly performance summary - Orange Light

Hello Orange Light team,

You recorded 49 bookings and 256,000 in revenue. Both results were below the hotel group benchmarks of 62.5 bookings and 285,500 in revenue.

Recommended next step: Review availability and pricing for the coming two weeks, then focus promotions on the channels that produced the most recent bookings.

Safety and operational controls

1. **Test mode first**
   Every email is redirected to one test inbox. Production recipients are enabled only after content review.

2. **No silent invention**
   Classification and numbers are calculated before the model call; the prompt prohibits unsupported explanations.

3. **Batch resilience**
   A malformed or unavailable AI response falls back to a validated template instead of stopping all hotels.

4. **Credential separation**
   Google, OpenAI, and Gmail secrets remain in n8n credentials or environment variables, never in the workflow files.

5. **Human review**
   The first scheduled run remains unpublished until the manual test output and recipient mapping are approved.

Verification and handoff

The reference implementation was run against the supplied Excel workbook in dry-run mode. It processed all 10 rows, calculated the expected medians, created 10 message previews, and redirected all recipients to the configured test address. No live emails were sent.

Reviewer setup

1. **Import**
   Import hotel_weekly_report_n8n.json into n8n.

2. **Connect**
   Select Google Sheets, OpenAI, and Gmail credentials in the matching nodes.

3. **Configure**
   Set the spreadsheet ID, sheet name, workflow timezone, and your test email.

4. **Test**
   Run the Manual test trigger and review all messages in the controlled inbox.

5. **Publish**
   After approval, turn off test mode, validate owner emails, and publish the Monday schedule.

Known limitations

- The source has no explicit reporting week, targets, prior-period values, room inventory, or currency metadata.
- Several mock recipient addresses use example.com and are intentionally unsuitable for a live send.
- Benchmark performance is relative to this 10-hotel portfolio; it does not yet measure each property's own commercial plan.
- A production rollout should add delivery logging, bounce handling, duplicate-run protection, and alerting for rejected rows.

Suggested production backlog

Priority	Change	Outcome
1	Add week, revenue goal, and order goal columns	Target-based scoring and historical audit trail
2	Write send status and message ID to a log sheet	Traceability and duplicate prevention
3	Add prior-week and four-week comparisons	Trend-aware messages instead of one-week snapshots
4	Add exception notification for invalid rows or failed sends	Faster operational recovery

References

Provided materials: AI Management Trainee Assignment.pdf and N8N Mock Data.xlsx.

Documentation: OpenAI Structured Outputs; n8n Schedule Trigger; Google Sheets Get Row(s); and Gmail Send.